

ANOUK VAN ALPHEN

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I'm a Bachelor of Computing student specializing in Software Engineering, working as a full-stack developer with a strong focus on C# backend development.

I bring structured insight, creativity, and balance to teams. My technical interests include cloud computing, networking, AI and machine learning, database transaction management, and linear programming optimization.

I have a background in classical ballet, contemporary dance and a passion for visual arts – this artistic background adds a unique and creative perspective to how I approach problem-solving and design.

I value collaboration and clear communication. My leadership style is hands-on, supportive, and centered on driving both the personal and professional growth of my team.

EXPERIENCE

Tutorly - Peer-Powered Learning Platform

Sept 2025 – October 2025

Developed a full-stack peer-to-peer tutoring platform for Belgium Campus students to schedule sessions with verified tutors, access an AI-powered chatbot, and participate in collaborative discussions. Project completed as part of the Software Engineering module.

[view the recreated GitHub here.](#)

Key contributions:

- Backend Development & System Architecture: Built APIs for authentication, tutor matching, and real-time messaging using ASP.NET Core, Blazor WebAssembly (.NET 8), and SignalR.
- Database & Storage Integration: Configured Supabase for relational data storage and Azure Blob Storage for large resources and AI knowledge base.
- AI Integration: Developed a RAG-powered chatbot, enabling context-aware academic support using retrieval-augmented generation, a python embedding service and calculated confidence scores to prevent hallucinations.
- Project Management: Led team using Agile Scrum methodology in Jira, coordinated sprints, and oversaw documentation, QA, and deployment planning.
- Impact: Delivered a fully functional platform supporting real-time collaboration and student-tutor matching, complete with additional gamification features and a comprehensive deployment plan.

Tech Stack: ASP.NET Core, Blazor WebAssembly, SignalR, Supabase, Azure Blob Storage, Python, Jira, Agile Scrum.

IoT-Based Water Quality Monitoring & Hyacinth Growth Prediction Project

Feb 2025 - May 2025

Developed a dashboard for monitoring water quality and predicting the spread of invasive water hyacinth in Hartbeespoort Dam as part of an international collaborative project between Belgium Campus ITVersity and PXL University. The system integrates IoT sensor data with machine learning models for predictive analytics. [view my project here](#)

Key Contributions:

- Backend & Data Pipeline: Processed and stored historical water quality data in Supabase, enabling efficient retrieval for analysis.
- Machine Learning Integration: Developed predictive models in Python to forecast hyacinth growth and environmental trends such as weather data.
- Frontend Development: Built a containerized dashboard using Vue.js and Vite for real-time data visualization and monitoring.
- System Deployment: Containerized backend and frontend services with Docker for consistent cross-environment deployment.
- Built a prototype using ESP32 microcontroller and temperature sensors to monitor water conditions in real time.
- On-the-Job Learning: Learned Python and adapted new tools to integrate ML models with cloud storage and dashboard functionality.

Tech Stack: Python, Vue.js, Vite, Supabase, Docker, IoT Sensors, Machine Learning

Student Council President

2025

As President of the Student Council at the Belgium Campus ITVersity Stellenbosch campus, I take a hands-on and highly involved approach to leadership. I work closely with my team to ensure strong collaboration, open communication, and personal growth for each member.

Some of my responsibilities include:

- Leading and facilitating weekly student stand-up meetings
- Actively supporting my team in their individual responsibilities and initiatives
- Organizing and delegating tasks while remaining involved throughout execution
- Acting as a liaison between students and faculty to address concerns and ideas
- Creating opportunities for leadership development within the team
- Fostering a vibrant, inclusive, and engaging student life

Read more about some of my student council initiatives:

[How I Enhanced Student Council Efficiency with Jira](#)

Other Projects & Technical Interests

- Strong foundation in C# backend development
- AI and machine learning
- Database transaction management and optimization
- Linear programming
- Exploring networking concepts, including LPR algorithms for packet routing (Dijkstra's algorithm) and optimizing serializable histories in distributed databases
- Research and experimentation

A wide range of projects, including group projects and exercises, can be found on my github.

Management Skills Training (Udemy)

As part of my ongoing effort to grow as an effective leader, I am currently enrolled in the "Management Skills Training for New and Experienced Managers" course on Udemy. This course is helping me develop core management competencies, including team motivation, communication, delegation, and conflict resolution. I'm actively applying these skills in my role as Student Council President, where I lead a diverse team to plan and execute campus initiatives.

EDUCATION

2017-2021

NSC, DF MALAN HIGH SCHOOL

2022

PROGRAMMING FOUNDATION, CTU TRAINING SOLUTIONS

With Python and SC-900 international qualifications.

2023-2025

BACHELORS IN COMPUTING, BELGIUM CAMPUS ITVERSITY

2025-2026

SOFTWARE ENGINEERING HONOURS, BELGIUM CAMPUS ITVERSITY

TECHNICAL SKILLS

- Programming & Development: C#, .NET, Java, JavaScript, HTML & CSS, SQL
- Full-Stack & Cloud: Building projects that integrate frontend and backend technologies
- Tools & Methodologies: Version control (Git), Jira project management, Agile teamwork exposure
- Optimization: Linear and non-linear programming

FRAMEWORKS, TOOLS & API'S

- Frontend & Backend: Vue.js, Node.js, Express.js, Tailwind CSS, Blazor WebAssembly
- Databases & Cloud: MongoDB, Supabase, Azure Blob Storage
- Development & Testing: Postman, Docker
- Hardware & IoT: Arduino, ESP32, Tinkercad

INTERPERSONAL & LEADERSHIP SKILLS

- Creative and logical problem-solving, pattern recognition, process improvement
- Strong conceptual and systems thinking
- Ethical decision-making and accountability
- Organized, dependable, and able to navigate complexity and ambiguity
- Effective communicator and active listener, skilled in presenting ideas with clarity

- Team player with a growing leadership mindset
- Comfortable working independently or taking charge when needed
- Adaptable to new technologies and challenges

SUBJECTS COMPLETED

- Linear Programming (181, 281, 381)
- Software Engineering
- Machine Learning
- Software Engineering (SEN381) – in progress
- Programming in C# (181, 182, 281, 282)
- Programming in Java (381)
- Web Programming (181, 281, 381)
- Mathematics (181, 281)
- Network Development
- Project Management (281, 381)
- Research Methods
 - With a research proposal on “Dynamic resource allocation in a multi-cloud environment”
- Computer Architecture and Design
- Information Systems (181, 281)
- User Experience Design
- Database Development (181, 281, 381)
- Entrepreneurship
- Innovation and Leadership (INL101, INL102, INL201, INL202)
- Internet of Things
- Software Analysis & Design
- Statistics (181, 281)
- Academic Writing

LANGUAGES

- Afrikaans: Native or bilingual proficiency
- English: Native or bilingual proficiency
- Korean: Limited working proficiency (in progress)

LINKS

[LINKEDIN](#)

[GITHUB](#)